PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : LIQUID SOLUTION (Question Bank # 08)

MM : 77 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. A mixture of A and B in the molar ratio 1 : 2 forms a maximum boiling azeotrope. Identify the incorrect statement, if A is more volatile [Molar mass of A = 100, Molar mass of B = 50]

(A) A liquid solution of A & B having mass % of A = 50 will have vapours having mass % of A = 50.

(B) A liquid solution A and B having mass % of A > 50% will have vapours having mass % of A > 50%.

(C) A mixture of A & B in the molar ratio 1 : 3 can be separated into azetropic mixture and pure A.

(D) A mixture of A and B in the molar ratio 2 : 3 can be separated into azeotropic mixture and pure A.

2. Air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298K, if the Henry's law constants for oxygen and nitrogen are 3.30×10^7 mm and 6.51×10^7 mm, respectively, the mole ratio of gases (O_2 : N_2) dissolved in water is about

(A) 1 : 2 (B) 2 : 1

(C) 1 : 8 (D) 1 : 3

3. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution at 100ºC is -

(Take K_b=0.76 K kg mol^{-1})

(A) 724 (B) 740

(C) 736 (D) 718

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4. 20 moles of liquid A are mixed with 20 moles of liquid B to form an ideal binary solution. Calculate moles of A in liquid state when half of the solution has vaporized.

(Given : $P_A^0 = 100\,\text{torr}, P_B^0 = 121\,\text{torr}$)

(A) $\frac{11}{21}$ (B) $\frac{110}{21}$ (C) $\frac{200}{21}$ (D) $\frac{220}{21}$

5. The freezing point depression of a 0.10 M solution of formic acid is -0.2046°C. What is the equilibrium constant for the reaction at 298 K?

$$HCOO^-(aq) + H_2O(l) \rightleftharpoons HCOOH\ (aq) + OH^-(aq)$$

(Given : $K_f(H_2O)$ = 1.86K kg mol^{-1} , Molarity = molality)

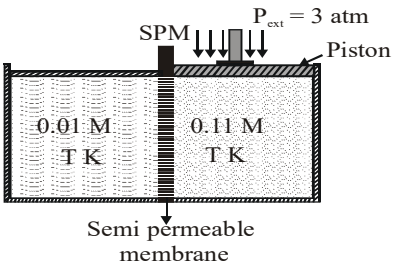
(A) 1.11×10^{-3} (B) 9×10^{-12} (C) 9×10^{-13} (D) 1.1×10^{-11}

6. 1kg of water is mixed with Sucrose and is cooled and maintained at –4°C. How much ice will be separated out if the moles of the sucrose is 0.75 in the solution ? $K_f(H_2O)$ = 1.86 Kg mol^{-1}K.

(A) 578.5 gm (B) 534.25 gm (C) 651.25 gm (D) 561.25 gm

7. A total pressure of 3 atm when applied on the surface of solution of concentration 0.11 M prevent flow of solvent across semipermeable membrane from other side of solution of concentration of 0.01 M, as shown in the diagram. What is the experimental temperature ?

(Take : R = 0.08 L-atm/K-mole)



(A) 375 K (B) 250 K (C) 125 K (D) 225 K

8. 1 litre of urea solution (1M) is mixed with 2 litre of 0.3M NaCl solution both at TK, osmotic pressure would be -

(A) 0.73 RT (B) 0.8 RT (C) 0.65 RT (D) 0.5 RT

9. Which of the following gas has lowest K_H for dissolution in H_2O at 25°C

(A) SO_2 (B) Ne (C) Ar (D) Kr

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(ii) One or more options correct Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

10. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult law. The normal boiling point (T_b) of pure acetone is less than that of pure CS_2. Pick out the **incorrect** statement(s) among the following.

(A) Boiling temperature of mixture is always less than boiling temperature of acetone.

(B) Boiling temperature of Azeotropic mixture is always less than boiling temperature of pure CS_2.

(C) When a small amount of CS_2 (less volatile component) is added to excess of acetone boiling point of resulting mixture increases.

(D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation.

11. Blood cell s in the human body have semipermeable membrane & depending upon concentration of solution inside blood cells & outside (in the blood); " Lysis" [expansion of blood cells] & " Crenatin"[Contraction of blood cells] may occur. Kidneys are responsible for keeping solution inside blood cell & blood at same concentration. Identify the options which is /are correct?

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(A) Lysis will occur when blood cells are kept in a solution which is isotonic with blood

(B) Crenatin will occur when blood cells are kept in a solution which is hypertonic with blood.

(C) Blood cells will have normal shape when placed in a solution isotonic with blood.

(D) Lysis occurs when blood cells are kept in hypotonic solution as compared to blood.

12. For an ideal binary liquid solution with $P_A^\circ > P_B^\circ$, which relation between X_A (mole fraction of A in liquid phase) and Y_A(mole fraction of A in vapour phase) is correct?

(A) $Y_A < Y_B$

(B) $X_A > X_B$

(C) $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$

(D) $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$

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13 A non - volatile solute 'X' completely dimerises in water, if the temperature is below–3.72°C and the solute completely dissociates as $X \rightarrow Y + Z$, if the temperature is above 100.26°C. In between these temperture,(including both temperature) 'X' is neither dissociated nor associated. One mole of 'X' is dissolved in 1 kg water.

Given: K_b of water $= 0.52$ K. Kg mol^{-1}

K_f of water $= 1.86$ K. Kg mol^{-1}

Which of the following statement(s) is/are **true** regarding the solution?

(A) The freezing point of solution is – 1.86°C

(B) the boiling point of solution is 101.04°C

(C) When the solution is cooled to – 7.44°C, 75% of water present initially will separate as ice.

(D) When the solution is heated to 102.08°C, 50% of water present initially will escape out as vapour.

(iii) Paragraph Type

This section contains **2 paragraphs** , describing theory, experiment, data etc. **6 questions** relate to 2 paragraphs with 3 questions. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

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Paragraph for question no. 14 to 16

The lowering of vapour pressure on adding a non-volatile solute may be measured by Ostwald-Walker method, In this method, dry air is first passed through a series of vessels having the solution, then through a series of vessels having pure solvent and finally through a vessel (normally U-tube) having the absorbent of the solvent. The masses of solution and pure solvent decreases due to removal of vapours of solvent in the flow of air and the mass of absorbent increases due to absorption of the vapour of solvent, in order to make the air dry. By measuring the changes in mass, we may determine the lowering of vapour pressure and hence, the composition of solution, form Raoult['s law.

14. If the mass of absorbent increased by 0.24 gm and the mass of pure solvent (water) decreased by 0.02 gm , then the mass percent of solute (glucose) in its aqueoue solution is :

(A) $\frac{1000}{21}\%$ (B) $\frac{100}{12}\%$

(C) $\frac{1000}{22}\%$ (D) $\frac{100}{11}\%$

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15. If the experiment is performed with aq. $AlCl_3$ solution ($\alpha = 0.8$) prepared by dissolving 1 mole $AlCl_3$ in 17 mole water and the decrease in mass of solution is found to be 0.18 gm, then the increase in mass of absorbent should be :

(A) 0.216 gm

(B) 0.225 gm

(C) 0.191 gm

(D) 1.08 gm

16. If the arrangement of vessels is changed in the experiment as :

Dry air $\xrightarrow{\text{Pure solvent}}$ $\xrightarrow{\text{solution}}$ $\xrightarrow{\text{Absorbent}}$ Dry air.

then, which of the following must be incorrect?

(A) the mass of pure solvent will decrease.

(B) the mass of solution will decrease.

(C) the mass of absorbent will increase.

(D) The experiment will obey the law of conservation of mass.

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Paragraph for Q.17 to Q.19

One mole of liquid 'A' and two moles of liquid 'B' are mixed 27ºC to form an ideal binary solution.

(ln2 = 0.7, ln3 = 1.1)

17. The value of ΔH_{mix} is

(A) 0

(B) + ve

(C) – ve

(D) any of these

18. The value of ΔG_{mix} is

(A) 0

(B) + 1140 cal / K

(C) – 1140 cal / K

(D) – 380 cal / K - mol

19. The value of ΔS_{mix} is

(A) 0

(B) + 3.8 cal / K

(C) – 3.8 cal / K

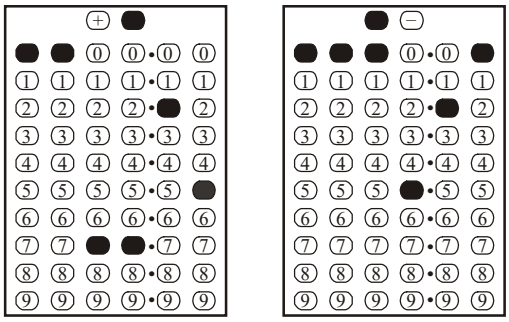
(D) + 3.8 cal / K - mol

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SECTION-II: (Integer Second Decimal)

- This section contains **4** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

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1. If 1 gm of solute dimerise upto 75% in 100 gm of H_2O & show depression in freezing point 0.093°C. Find average molecular weight of solute in the solution $\left(K_{f(H_2O)} = 1.86\right)$.

2. 4ml of a gas at 1 atm pressure and 300 K is dissolved in 1L of solution. Calculate the volume of the gas dissolved at 4 atm pressure and 300 K in $\frac{1}{2}$ L of solution **in ml**.

3. In how many of the following, the elevation in boiling point of the final solution is more than that of initially taken solution (No limiting reagent) (Assume $\alpha = 1$ where ever needed)

(i) $BaCl_2(aq.) + Na_2SO_4(aq.)$ (added)$\rightarrow BaSO_4 + 2NaCl(aq.)$

(ii) $K_4Fe(CN)_6(aq.) + 2CuSO_4(aq.)$ (added) $\rightarrow Cu_2Fe(CN)_6 + 2K_2SO_4$

(iii) $Al_2(SO_4)_3(aq.) + 3BaCl_2$ (aq.) (added) $\rightarrow 3BaSO_4 + 2AlCl_3(aq.)$

(iv) $AgNO_3$ (aq.) + 2KCN (added) $\rightarrow K[Ag(CN_2)(aq.) + KNO_3(aq.)$

4. When the mixture of two immicible liquids (water and nitrobenzene) boils at 372 K and the vapour pressure at this temperature are 98.4 kPa (H_2O) and 3.6 kPa ($C_6H_5NO_2$). If the weight ratio of nitrobenzene and water in the vapour is 1 : x, then the value of 'x' is.

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ANSWER KEY

SECTION–I : (i) Only One option correct Type : 3(–1)

1.	Ans.(C)	2.	Ans.(A)	3.	Ans. (A)	4.	Ans.(D)
5.	Ans.(B)	6.	Ans.(C)	7.	Ans.(A)	8.	Ans.(A)
9.	Ans.(A)						

(ii) One or more options correct Type : 4(–1)

10.	Ans. (A, C, D)	11.	Ans.(B,C,D)	12.	Ans. (C)	13	Ans.(A,B,D)

(iii) Paragraph Type : 3(0)

14.	Ans.(A)	15.	Ans. (A)	16.	Ans. (B)

Paragraph for Q.17 to Q.19

17.	Ans.(A)	18.	Ans.(C,D)	19.	Ans.(B)

SECTION-II: (Integer Second Decimal) : 4(0)

1.	Ans.(200.00)	2.	Ans. (2.00)	3.	Ans.(4.00)	4.	Ans.(4.00)

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